Deep Fake Detection

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Introduction

Data Science is an increasingly growing field. The applications of Data Science can be used in nearly every other field. This project sought to apply the tools of Data Science to solving the problem of deep fake images of people. First, the OpenForensics Dataset was used to provide a large quantity of labeled images to facilitate a supervised learning method. Then, the photos were cropped with the cv2 library in Python so only the faces in the photos would be analyzed. The photos were then normalized to a predetermined aspect ratio before training. Finally, the model was trained on the cleaned training portion of the dataset and tested and validated on their corresponding datasets.

Related Work

Previous studies have provided foundations for real and fake image classification from different perspectives. Rössler et al. constructed the FaceForensics++ dataset and demonstrated that CNN-based methods can effectively detect multiple types of facial manipulation images, providing an important benchmark for supervised deep fake detection. Nataraj et al. found that GAN-generated images can alter pixel-level statistical features. Therefore, they used co-occurrence matrices extracted from RGB channels together with a CNN for detection, achieving high classification accuracy on the CycleGAN and StarGAN datasets. Xuan et al. focused on the generalization problem and pointed out that detectors can easily rely on unstable low-level noise features. They found that preprocessing methods such as Gaussian Blur and Gaussian Noise can help the model learn more generalizable differences between real and fake images. Wang et al. further showed that a classifier trained only on ProGAN images can generalize well to many unseen CNN-generated images after using data augmentation such as blur and JPEG compression. Chai et al. also found that fake traces are often concentrated in local regions, such as hair, background, and facial details. Therefore, a patch-based classifier can help explain the basis on which a model identifies fake images, while also reminding us that preprocessing differences may lead to misleadingly high generalization performance. Ojha et al. further pointed out that traditional real-vs-fake classifiers can easily fail when facing newer generative models such as diffusion models, and demonstrated that nearest neighbor or linear probing methods based on CLIP-ViT features have stronger cross-generative-model generalization ability.

Methodologies

Data Cleaning

The data was multiple sets of photos that were wrapped in a zip file. The methodology used opened the zip file in the program ensuring any user that recreates this work will not be required to purchase a program to open zip files. The folder paths for each dataset and their two categories, real or fake, were saved in variables. A function was created that opened the zip file and followed a provided path to extract a desired set of photos. The photos were copied into grayscale then processed through a face recognition function provided by the cv2 library in Python. The face recognition function returns the bounds of the face found, so the original function can create a cropped copy of the original photo around only the face found. Photos with several faces would generate several copies that were all placed in a list. This list would then be iterated through to save the photos to the clean data folder to be organized the same way as the raw data. The photos were also transformed into numpy arrays for the model to perform training and analytics on.

Data Augmentation

Base Transforms

For training, validation, and testing, the images are transformed using a custom resize and pad function. If the image's width or height is greater than 150 pixels, the image will be resized to a maximum long edge of 150 pixels. The image's aspect ratio is maintained during the resizing process. After resizing, if there is extra space, padding will be added to the image to fill a 150 by 150 pixel space. The image is centered in the middle of the padding space. The images are then converted to tensors and normalization is applied.

Augmentation Transforms

In addition to the base transforms, augmentation transforms were applied only to the training data. These additional augmentations included random horizontal flip at 50% probability and random color jitter of brightness, contrast, and saturation at 20% each.

Model Structure

The model is set up using PyTorch. We used the ResNet50 as the model feature extractor. The final classification layer of the ResNet50 is removed. We added our own classification head to the model consisting of 2 linear layers separated by a ReLU function and a dropout rate of 0.2. The final output of the model is the raw logit.

Training

The dataset provides predetermined subsets for training, validation, and testing. The model was trained using the training subset provided by the dataset. Every epoch validation was completed reporting loss and accuracy on the validation subset of the data. At the end of training a single check was performed on the test subset of the data analyzing loss and accuracy.

The following hyperparameters were used for training.

Hyperparameter	**Value**
num_epochs	30
batch_size	128
num_workers	2
Patience	5
min_delta	0.0005
learning_rate	0.0001
betas	(0.9, 0.990)
weight_decay	0.0001
setp_size	7
gamma	0.5
criterion loss function	BCEWithLogitsLoss
optimizer	AdamW
learning rate scheduler	StepLR

The training loop had a check at the end of each epoch. The check was reviewing the validation loss. It saves the new best model if the performance was better than previous versions within the constraint of the min_delta hyperparameter. If a better model can’t be achieved within the patience hyperparameter, the training will stop automatically.

Model Explanation Using Grad-CAM Heatmap

We used gradient-weighted class activation mapping to better understand what portions of the image the model is focusing on. We targeted the final convolutional layer in the ResNet50 feature extractor. We create a class activation map as a weighted sum of feature maps. The heatmap is then normalized and scaled to match the original image. The heatmap is then overlaid on the original image to show the regions the model is focusing on.

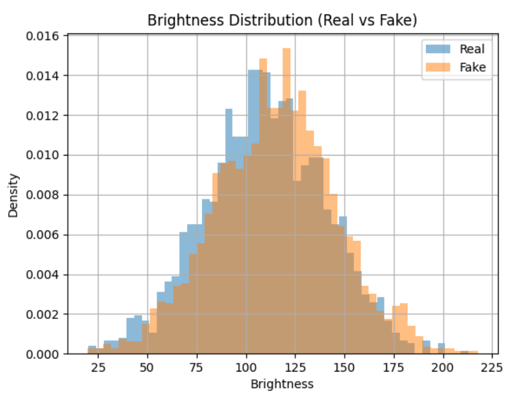
Results & Discussion

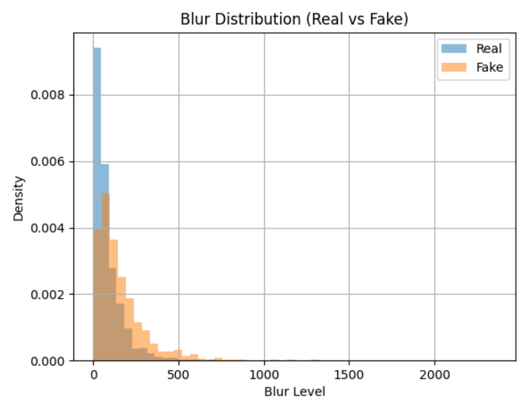
Exploratory Data Analysis

The dataset used for this project was the OpenForensics Dataset designed by Trung-Nghia Le, Huy H. Nguyen, Junichi Yamagishi, and Isao Echizen. The dataset was designed for generating face forgery detection and segmentation models. The deep fake images were real photos that were altered through various methods. In the fake photos, some of the faces are real, but one or two are fake; therefore, the model must be able to detect individual faces and determine if that singular face is real or fake (Le).

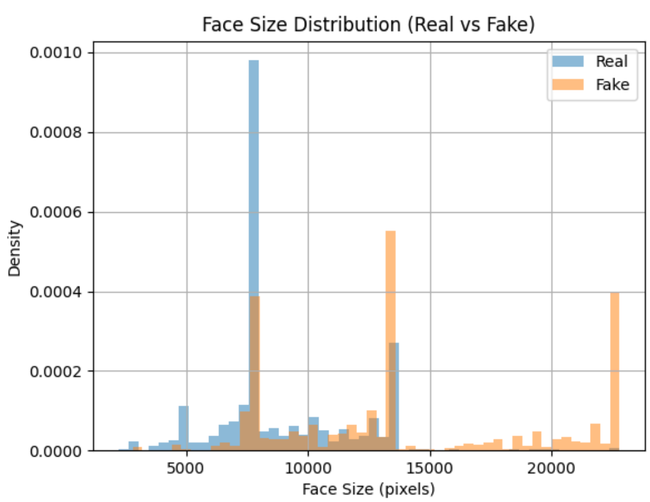
Additional Exploratory Data Analysis was conducted to better understand differences between real and fake images from a statistical and visual perspective. This section focuses on evaluating handcrafted image features to understand their effectiveness in distinguishing real and fake images. All analyses were conducted using the same randomly sampled subset of images to ensure consistency and fair comparison across features. OpenCV is used to process images and detect faces using a pre-trained Haar Cascade model, this allows us to locate faces in each image and extract important features such as face size and number of faces, which are essential for our analysis.

First, brightness and blur features were extracted from randomly sampled images in both classes. The results showed that while brightness values were relatively similar between real and fake images, blur values were significantly higher in fake images, suggesting that fake images tend to have lower visual sharpness and higher quality inconsistency.

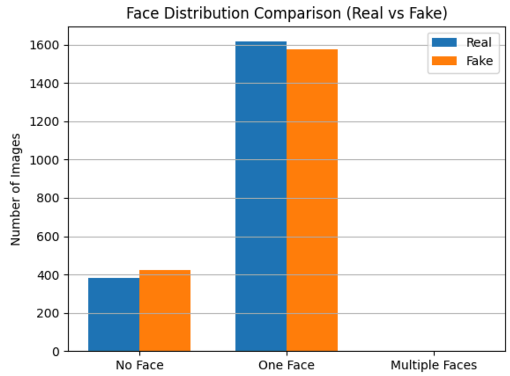




Second, face size analysis was performed by measuring the bounding box area of detected faces. Fake images showed a higher average face size and greater variability compared to real images, indicating potential scaling or generation artifacts in synthetic faces.



Finally, face count distribution was analyzed for both classes. The results showed no significant difference between real and fake images in terms of the number of faces present, suggesting that this feature alone is not a strong discriminator.

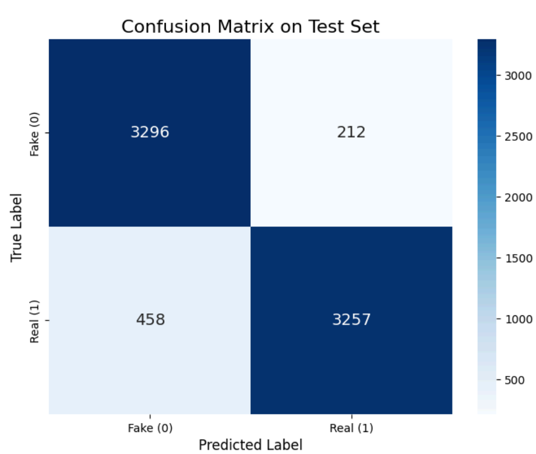


Overall, these findings indicate that image quality-related features (such as blur and face size) are more informative for distinguishing real and fake images than simple structural features such as face count.

Grad-CAM Heatmap Findings

Grad-CAM is a technique that generates a heatmap showing the regions of an image that have the most influence on a convolutional network's prediction. The Grad-CAM technique was used to generate headmaps of the model inference to better understand the internal workings of the model. For real images, the model tends to focus on the central facial regions. However, for deep fake images, the model tends to focus on smaller points closer to the edges of the images that may not be part of the face. This could indicate that the model has learned some shortcut to predicting the image class. Below are several examples of the heatmaps.

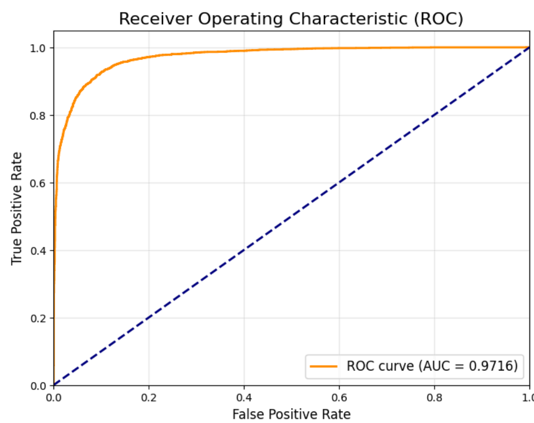
Confusion Matrix



In this part of the model evaluation, Real images are treated as the positive class, while Fake images are treated as the negative class. Therefore, Real images correctly predicted as Real are counted as True Positives, and Fake images correctly predicted as Fake are counted as True Negatives. After data cleaning, the test set contained 7,223 images in total, including 3,508 Fake images and 3,715 Real images. The two classes were relatively balanced, with no severe class imbalance. The confusion matrix shows that the model correctly classified 3,296 Fake images and 3,257 Real images, resulting in an Accuracy of 90.72%, which indicates that the model achieved strong overall classification performance on the test set.

With Real defined as the positive class, the model's Precision was 93.89%. This means that among all images predicted as Real, most were indeed Real images, so the model's Real predictions were generally reliable. The model's Recall was 87.67%, indicating that some Real images were still not successfully identified. In addition, the model's Specificity for the negative class, Fake images, was 93.96%, showing that the model was also effective at identifying Fake images. The confusion matrix also reveals two types of classification errors: 212 Fake images were incorrectly predicted as Real, which are Type I Errors, while 458 Real images were incorrectly predicted as Fake, which are Type II Errors. The number of Real images misclassified as Fake was clearly higher than the number of Fake images misclassified as Real. This suggests that, among the classification errors, the model more frequently misclassified Real images as Fake, reflecting that it still has room for improvement in fully identifying all Real samples. Overall, the confusion matrix indicates that the model has strong deepfake classification performance, but the relatively high number of False Negatives suggests that future work could reduce Real images being misclassified as Fake by adjusting the classification threshold or further optimizing the model.

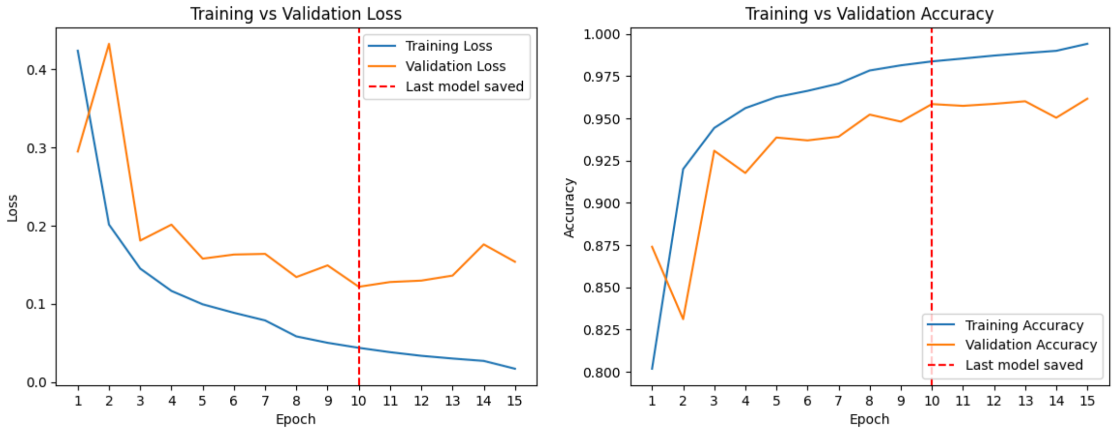
ROC Curve



In the ROC curve analysis, Real images are still treated as the positive class, while Fake images are treated as the negative class. The ROC curve shows the relationship between the True Positive Rate and the False Positive Rate under different classification thresholds. Therefore, compared with the confusion matrix, which is based on a single threshold, the ROC curve provides a more comprehensive evaluation of the model's ability to distinguish Real images from Fake images. As shown in the figure, the ROC curve is clearly above the diagonal dashed line, which represents a random classifier, and the curve is also very close to the upper-left corner. This indicates that the model can achieve a high True Positive Rate while maintaining a low False Positive Rate. The AUC value of the model is 0.9716, which is close to 1.00, suggesting that the model has strong discriminative ability between Real and Fake images. In other words, the model not only performs well under the selected classification threshold, but also maintains strong overall performance across different threshold settings.

Model Accuracy & Metrics

The model achieved a loss of 0.12 and an accuracy of 95.8% on the testing dataset. The following plots show loss over epochs and accuracy over epochs respectively.



Some extra data visuals were generated to better understand the efficiency of the model, some errors in the model, and insight into the parameters of the model. In order to better understand the efficiency of the model, a probability distribution histogram was generated to visualize how well the model separated the two classes. The probability distribution histogram shows clear separation between the real and fake class with the majority of each class being on the edges. A threshold sweep curve was generated with the idea of maximizing our accuracy, precision, and recall. A threshold score of around 0.25 would maximize the three stats; however, higher threshold values could be utilized to gain more precision at approximately the same accuracy at the expense of a lower recall score. Finally, the images of people's faces that were misclassified were compiled to understand what caused our model to falsely identify them. Poor picture quality, shading around key features, and bad image crops are easily identifiable culprits for these misclassifications.

Conclusion & Future Work

Our model was able to successfully detect deep fake images with high accuracy and strong discrimination between classes. However, the model did still struggle with a few difficult samples. Our future work includes further exploring the capabilities of our current model, exploring the Grad-CAM heatmap results to better understand why the model focuses on different points in real vs fake images, and further enhancing the model structure to achieve stronger identification on the difficult samples.

This project successfully implemented a deep fake detector to detect if an image is generated or real. We explored the image dataset for better understanding. We implemented a deep learning model for binary classification of images. We explored and explained the capabilities of the model using several techniques. This model could be used for identification of deep fake images to discover abuse of AI image generation.

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